



A biocompatible and recyclable ionic liquid platform for long-term and high temperature plant RNA stabilization

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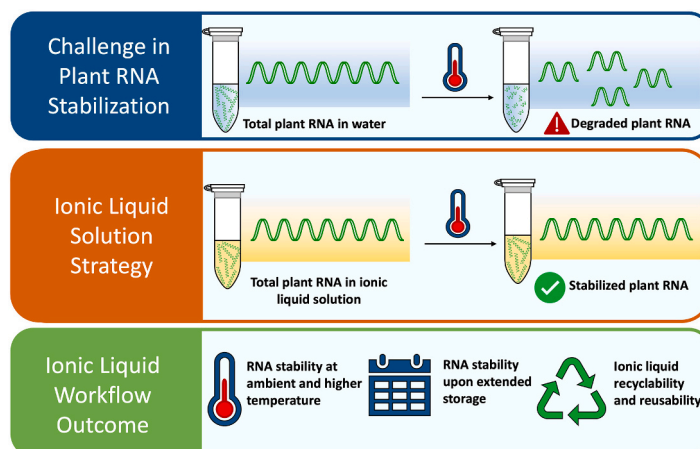
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HIGHLIGHTS

- Identified 20% (w/v) choline glutamate as a biocompatible plant RNA stabilization medium.
- Stability demonstrated at ambient and higher temperatures.
- Developed a rigorous analytical framework for validating RNA quality.
- IL recycled and reused for 5 cycles without loss of RNA purity or integrity.
- Provided an affordable, safe, recyclable solution for RNA storage and transport.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: RNA is a fundamental biomolecule essential to life with diverse and expanding functional roles. The inherent instability and high susceptibility of RNA to degradation during sampling, handling, and storage limit the reliability of RNA-based analyses. Elevated temperature conditions accelerate degradation and compromise RNA quality for downstream applications. Ionic liquids (ILs) have shown significant promise in stabilizing short-chain and commercially-available RNA. There is a critical need for stabilization platforms that protect more complex and heterogenous RNA systems, such as total plant RNA across a broad range of temperature conditions, while maintaining biocompatibility and enabling recyclability.

Results: In this study, choline-based aqueous IL solutions were evaluated to develop a biocompatible and reusable platform for stabilization of total plant RNA. Among the different ILs tested, 20% (w/v) choline glutamate

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emerged as the most effective stabilization medium which hindered RNA degradation during short-term handling at higher temperature and extended storage at 40 °C and ambient temperature. Immediate stabilization preserved RNA integrity at levels statistically comparable to RNA controls. Upon heating, control RNA degraded at 70 °C and higher, whereas RNA stored in choline glutamate IL retained higher integrity even at 80–90 °C, indicating significant suppression of thermal degradation. Long-term studies demonstrated higher RNA integrity for at least 72 h at 40 °C and delayed degradation by several weeks at room temperature. The IL was recycled and reused for at least five cycles without compromising RNA integrity or purity, representing a significant advancement in green analytical chemistry.

Significance: This work introduces an IL-based platform specifically designed for stabilizing complex, heterogeneous total plant RNA. By identifying choline glutamate as an effective medium, this study advances the field toward a practical, scalable preservation strategy that operates across a wide temperature range and supports long-term storage, including conditions relevant to field sampling and transport. The integration of solvent recyclability while maintaining RNA integrity further enables closed loop, low waste workflows applicable to complex biological samples.

1. Introduction

RNA is a ubiquitous biopolymer found in all prokaryotic and eukaryotic cells and has emerged as a fundamental biomolecule in modern biology due to its diverse and expanding functional roles, encompassing not only information transfer from DNA to proteins [1] but also regulatory, catalytic, and structural activities [2]. RNA can provide more information that is not directly evident from the genome, as RNA transcripts offer insight into gene regulation and protein-coding activity and, unlike the largely static genome, reflect genes that are actively expressed at a given time and are responsive to external conditions [3]. Ongoing research in the field of RNA biology and chemistry has resulted in significant advancement in the development of RNA-based drugs for therapeutic applications [4]. A wide range of these bioanalytical, diagnostic, and therapeutic technologies depend on high-quality RNA. However, the inherent thermal and enzymatic instability of RNA presents a significant challenge, establishing RNA quality and integrity as a critical issue in modern biological research [5]. RNA is more sensitive to degradation than DNA due to the structural instability arising from the 2'-hydroxyl group which can cleave its neighboring phosphodiester bond by in-line nucleophilic attack on the 3'-vicinal phosphate [6]. This reaction can occur spontaneously and is accelerated under alkaline pH conditions and by heat [6]. RNA also undergoes degradation in the presence of contaminating nucleases. Maintaining RNA integrity throughout extraction, storage, and application is a critical requirement for reliable bioanalytical performance and biological efficacy.

In an effort to decrease degradation, RNA is typically stored in nuclease-free water at low temperatures (−20 °C or −80 °C), or in liquid nitrogen for long-term preservation. Transporting RNA samples is typically performed on dry ice, which is expensive and often poses logistical challenges for air transport due to regulatory restrictions, weight limitations, and long-distance travel constraints [7]. To overcome many of the challenges in RNA storage, several matrices have been developed including GenTegra RNA, RNastable, and RNashell. GenTegra RNA is a water-soluble inert chemical matrix that preserves RNA integrity in the dry-state for storage and transport by protecting samples from hydrolysis and oxidation [8]. GenTegra RNA provides excellent RNA stabilization, but sample drying is required prior to storage or shipping. The drying process typically takes 10–16 h using a FastDryer, ~2–4 h using a vacuum desiccator or SpeedVac and ~18–24 h in a biosafety hood [9]. RNastable utilizes a similar process where RNA is desiccated in the presence of a stabilization matrix [10]. RNashell preserves RNA by vacuum drying in the presence of a stabilizing agent in stainless steel minicapsules [7]. Each minicapsule needs to be assembled in a controlled (anoxic, anhydrous) and inert (argon, helium) atmosphere and sealed by pulsed laser YAG welding [11]. To recover RNA, the welded minicapsules need to be disassembled using a shell opener. While these commercial products offer advantages in RNA stabilization, each has their own complexities, such as high operational complexity,

need for specialized tools for sample recovery and limited or no reusability.

Recently, ionic liquids (ILs) have been explored as a promising class of solvents for nucleic acid extraction from complex sample matrices [12,13], nucleic acid storage and stabilization [14–16], and for use in sustainable chemical research [17]. ILs are organic molten salts featuring melting temperatures at or below 100 °C [18] and desirable physicochemical properties such as negligible vapor pressures, high chemical stability, and high ionic conductivity. These characteristics, combined with the structural tunability of ILs owing to the broad range of cation and anion combinations, have enabled the rational selection of IL formulations that interact favorably with nucleic acids and suppress degradation. In this context, IL systems have demonstrated promise for RNA preservation. A hydrated buffered choline dihydrogen phosphate IL has been shown to substantially extend the shelf-life of small interfering RNA (siRNA) for up to three months, even in the presence of degrading enzymes such as RNase A [19]. While nucleases are catalytically active in water and degrade nucleic acids, choline dihydrogen phosphate creates a non-hydrolytic environment that inactivates nucleases [20]. Choline-based ILs containing anions derived from Good's buffers have been used for stabilization of recombinant small RNA [21]. Subsequent research has focused on the development of amino acid-based IL platforms for a combined small RNA (sRNA) extraction and preservation process from bacterial lysates; however, further improvements in RNA purity are required [22]. While ILs have shown significant promise in stabilizing short-chain and commercially available RNA, exploring IL-based strategies using ILs to longer and more complex RNA remains critical, as RNA stability is shown to decrease with increase in length of RNA [23]. In this regard, plant RNA represents a relevant and challenging model system as it comprises a heterogeneous mixture of various RNA types, including messenger RNAs (mRNAs), ribosomal RNAs (rRNAs), transfer RNAs (tRNAs), and smaller regulatory RNAs. Among these, 25S and 18S rRNA are considerably larger than small RNAs, represent the most abundant RNA, and are commonly used as indicators of RNA integrity [24]. Thus, exploring stabilization of total plant RNA using ILs requires not only protecting RNA against degradation, but also ensuring compatibility with downstream bioanalytical workflows.

To determine whether RNA stabilization is successful, its quality must be quantified in terms of purity and integrity using spectrophotometric, fluorescence-based, and electrophoretic methods. Full-spectrum ultraviolet (UV) absorption analysis using NanoDrop spectrophotometry with absorbance ratios (A260/280 and A260/230) provide information regarding RNA purity and the presence of contaminants. However, quantifying RNA by measuring absorbance at 260 nm can be unreliable and inaccurate as UV absorbance measurements are not selective and cannot distinguish between DNA and RNA [25]. Fluorescence-based quantification using Qubit fluorometer is more selective, sensitive, and accurate for quantitating nucleic acids [25]. RNA integrity is mostly assessed using gel electrophoresis or microfluidic lab-on-chip systems (e.g., Bioanalyzer), which visualize RNA profiles and evaluate degradation

through features such as 18S and 28S rRNA bands and their ratio, providing a standardized measure of RNA integrity. While the bioanalyzer is an automated device that reproducibly performs electrophoretic separations [26], agarose gel electrophoresis relies on human interpretation and is not comparable from one lab to another [27]. Despite the established use of these bioanalytical tools, the majority of studies reporting ILs as RNA stabilizers have validated the integrity and stability of RNA based on circular dichroism (CD), RNA melting temperatures, and gel electrophoresis. To date, ILs have not yet been explored for stabilizing longer and more complex RNA systems, such as plant RNA.

This study addresses a critical gap in RNA preservation by introducing the first IL-based platform tailored for stabilizing complex, heterogeneous total plant RNA. We establish 20% (w/v) choline glutamate as a biocompatible and recyclable medium and applied a standardized, multi-metric framework for evaluating plant RNA stability that integrates spectrophotometric, fluorescence-based, and microfluidic electrophoretic analyses. This combined methodological and materials innovation enables reliable RNA preservation across a wide range of conditions relevant to real-world applications, including elevated temperatures and long-term ambient storage. Furthermore, the demonstrated recyclability of the IL without loss of RNA integrity introduces a sustainable, closed-loop approach to nucleic acid preservation. Collectively, this work advances RNA stabilization toward scalable, field-deployable, and environmentally responsible workflows for plant molecular studies.

2. Materials and methods

2.1. Materials and reagents

Choline hydroxide (45 wt% aqueous solution) and L-glutamic acid were purchased from Thermo Scientific (Carlsbad, CA, USA). Methanol (ACS grade) was acquired from Thermo Fisher Scientific (Waltham, MA, USA). UltraPure™ DNase/RNase-Free distilled water was purchased from Invitrogen™. Phosphoric acid (85%) was purchased from Fisher Scientific. Ethanol 200 proof was purchased from Decon Labs, Inc (King of Prussia, PA, USA). All materials used in the experiments were RNase-free. An agate mortar (100 mm O.D. × 82 mm I.D. × 25 mm depth) with a pestle acquired from MSE supplies (Tucson, AZ, USA) was used for sample grinding. An Accumet AB-150 pH meter from Fisher Scientific was used for pH measurements.

2.2. Synthesis of ionic liquids

Choline-based ILs were synthesized according to previously reported procedures [22,28,29]. The exact concentration of 45 wt% choline hydroxide solution was determined by a titration with standardized hydrochloric acid prior to use. Choline formate was prepared by the dropwise addition of formic acid to choline hydroxide (1:1) in water and stirred at ambient temperature conditions overnight. Water was removed under reduced pressure, first using rotary evaporation at 50 °C, and then dried in a vacuum oven for 3 days.

Choline dihydrogen phosphate was prepared by the dropwise addition of phosphoric acid (1:1) and stirred at ambient temperature overnight. Water was removed under reduced pressure using rotary evaporation at 50 °C. The solid product was dissolved in methanol followed by reprecipitation by dropping in acetone. The reprecipitation process was repeated four times. The purified product was dried in a vacuum oven for 3 days.

Choline glutamate was synthesized by neutralization of choline hydroxide with glutamic acid (1:1) under stirring at ambient temperature overnight. The synthesized ILs were washed with a 7:3 v/v mixture of acetonitrile and methanol to remove unreacted amino acids. Excess solvent and water were removed under reduced pressure using rotary evaporation at 50 °C. The purified ILs were dried in a vacuum oven for 3

days. The ILs were characterized by ¹H NMR spectroscopy on a Bruker Avance NEO 400 MHz spectrometer (Billerica, MA, USA) and shown to be >97% pure. Samples were analyzed in deuterium oxide and the corresponding ¹H NMR spectra and chemical structures of the ILs are provided in Figs. S1–S3.

2.3. Plant RNA extraction

Plant RNA was extracted from soybean (*Glycine max* L. Merr) seeds using the RNeasy plant mini kit (QIAGEN), according to the manufacturer's instructions and quantified using a Qubit 4.0 fluorometer (ThermoFisher Scientific, Waltham, MA, USA) with the Qubit™ RNA Broad Range assay.

2.4. Quality assessment of RNA

The purity of recovered RNA was evaluated by a NanoDrop 2000c spectrophotometer (Thermo Scientific) and the A260/280 and A260/230 ratios were assessed. The integrity of recovered RNA was assessed using an Agilent 2100 Bioanalyzer System with the Plant RNA Nano and Pico assay in accordance with the RNA 6000 Nano and Pico kit protocol (Agilent Technologies, Waldbronn, Germany). Data analysis was performed on the 2100 Expert software, version B.02.12 2420 (SR1). The quality of RNA was assessed using the RNA integrity number (RIN), which is represented by a numerical score ranging from 1 (most degraded) to 10 (most intact) and the 25S/18S rRNA ratio was obtained from the electropherogram output generated by the software.

2.5. Aqueous solutions of ILs as plant RNA stabilization media

Plant RNA was dissolved in 10 µL of different ILs as an aqueous solution such that the final concentration of the IL was 20% (w/v) and the final concentration of RNA was 150 ± 10 ng/µL. The mixtures were incubated for 0.5 h at room temperature and RNA was precipitated by adding 2.5 vol of ethanol, incubating at –80 °C, and centrifuging at 15000 g for 1 h at 4 °C. The supernatant was removed and the RNA pellet was briefly air-dried at room temperature and reconstituted using 8 µL of RNase-free water. All experiments were performed in triplicate.

2.6. RNA thermal stability study

To evaluate the thermal stability of RNA in aqueous IL solutions, plant RNA was dissolved in each IL solution to achieve a final IL concentration of 20% (w/v) and an RNA concentration of 150 ± 10 ng/µL. Samples were then incubated for 30 min at room temperature, 40 °C, 50 °C, 60 °C, 70 °C, 80 °C, and 90 °C using a Bio-Rad CFX96 Touch real-time PCR thermocycler (Hercules, CA, USA). After incubation, each sample was immediately placed on ice and RNA was precipitated from the IL solutions, as described in the aforementioned procedure. All experiments were performed in triplicate.

To evaluate the thermal stability of RNA for extended periods of time, plant RNA was dissolved in 10 µL of 20% (w/v) choline glutamate IL solution and incubated at 40 °C for 3 h, 6 h, 12 h, 24 h, 48 h and 72 h and immediately placed on ice. RNA was precipitated from the IL solutions as described in the aforementioned procedure. Control experiments were performed with 150 ± 10 ng/µL of plant RNA dissolved in RNase-free water and subjected to the identical incubation conditions. All experiments were performed in triplicate.

2.7. Room temperature RNA stability study

To assess RNA stability at room temperature over an extended period of time, plant RNA (150 ± 10 ng/µL) was dissolved in 20% (w/v) choline glutamate IL solution and stored at room temperature. The RNA-IL mixture (10 µL) was removed every week, precipitated as described above, and reconstituted with 8 µL of RNase-free water. All experiments

were performed in triplicate.

2.8. Recycling and reuse of IL for plant RNA storage

Plant RNA (150 ng/ μ L) was incubated in 20% (w/v) choline glutamate IL solution (250–150 μ L) for 1 h at room temperature, precipitated (*vide supra*), and reconstituted with 24 μ L RNase-free water. The choline glutamate IL was recovered from the supernatant by removing solvent using rotary evaporation until a constant weight was achieved. The concentration of the recovered IL was adjusted with water to 20% (w/v) and reused for a new batch of plant RNA of the same concentration. This cycle was successfully repeated for 5 consecutive iterations. All experiments were performed in triplicate.

2.9. Circular dichroism spectroscopy

Circular dichroism (CD) measurements were carried out with a Jasco J-710 spectropolarimeter (Jasco, Easton, MD, USA) using a Peltier-type temperature control system. CD spectra were acquired at a constant temperature of 25 °C using a scanning speed of 20 nm/min, with a response time of 2 s over wavelengths ranging from 210 to 320 nm. The recording bandwidth was 1 nm with a data pitch of 0.2 nm using a quartz cell with an optical path length of 1 mm. Three scans were averaged per spectrum to improve the signal-to-noise ratio, and the spectra were smoothed by using the Savitsky-Golay algorithm with a convolution width of 25 points in the operating software. All spectra were subtracted using the blank.

CD melting experiments were performed in the temperature range from 10 to 95 °C, with a heating rate of 5 °C/min, by monitoring the ellipticity at 265 nm. Data were converted into fraction folded plots, as reported in literature [21] using equation (1),

$$\theta = \frac{CD_{\lambda} - CD_{\lambda}^{\min}}{CD_{\lambda}^{\max} - CD_{\lambda}^{\min}} \quad (\text{Equation 1})$$

where CD is the ellipticity of the monitored wavelength at each temperature, and $CD^{\min}$ and $CD^{\max}$ represent the lowest and highest ellipticity, respectively [21]. Data points were fitted to a Boltzmann distribution (OriginPro 2025) and the melting temperatures (T_m) were determined by taking the first derivative from the plots. The total RNA concentration used was 150 ± 10 ng/ μ L.

3. Results and discussion

3.1. Aqueous solutions of ILs as stabilizers for plant RNA storage

ILs have been explored as promising alternative RNA storage media that offer potential advantages over traditional methods, such as freezing, due to their ability to stabilize nucleic acids. Unlike purified siRNA or bacterial sRNA, total plant RNA extracts contain a heterogeneous mixture of various RNA types, including mRNAs, rRNAs, tRNAs and smaller regulatory RNAs. Exploring the stability of total plant RNA in different IL formulations aids in understanding whether an IL formulation protects a biologically and chemically diverse sample matrix (not just a single oligonucleotide), which is essential for applications in transcriptomics, diagnostics, biobanking and field sampling where intact, representative RNA is required. Many molecular biological applications such as RNA-sequencing, reverse transcription-quantitative polymerase chain reaction (RT-qPCR) and microarrays depend on intact RNA [30]. This study aims to identify IL chemistries that work best in stabilizing plant RNA by studying their optimal concentration, thermal stability, shelf-life at ambient temperatures, and any tradeoffs (e.g., RNA purity vs. stabilization). These findings enable the development of standardized, validated IL-based storage workflows for plant molecular research using RNA. To the best of our knowledge, studies exploring the stability of total plant RNA in aqueous IL solutions have

not yet been reported.

The initial stabilizing performance of three different IL formulations was assessed by incubating plant RNA from soybean seeds in 20% (w/v) IL for 0.5 h at room temperature followed by ethanol precipitation. RNA quality was measured by using the RIN and the 25S/18S rRNA ratio. The RIN is a measure of the integrity and level of degradation of a particular RNA sample, obtained from the Agilent Bioanalyzer, and is used to determine if an RNA sample is acceptable for sensitive downstream applications such as sequencing, microarray analysis, or RT-PCR [31]. The RIN takes into account the entire electrophoretic trace of an RNA sample, which allows for a more detailed analysis of RNA degradation patterns compared to traditional methods [27]. A higher RIN (closer to 10) indicates better RNA quality (i.e., less degradation). A ratio of the 25S to 18S ribosomal subunits [25 S/18S rRNA ratio] of ~ 2.0 is ideal for high quality RNA and this ratio decreases as RNA fragments [24]. The presence of high 18S and 25S rRNA peaks, as well as low 5S RNA, indicates intact RNA. A degraded RNA sample would exhibit a shift in the RNA size distribution toward smaller fragments and ultimately a decreased fluorescence signal [30]. Bioanalyzer electrophoretic patterns of plant RNA recovered from 20% (w/v) ILs reflect the general characteristics of high-quality plant RNA for the control, as shown in Fig. 1. The 5S region (peak 2) represents the small RNAs, such as 5S rRNA and tRNA [27]. The 18S (peak 6) and 25S (peak 8) rRNA fragments were clearly visible and prominent in all samples; other identifiable fragments were plastid rRNA peaks (peaks 4, 5, and 7) [32]. The remaining peaks occurred within the so-called “fast region” (region 3) between the small RNAs and 18S rRNA [27]. No distinct peaks were observed in the post region (the region past the rRNA peaks). Electropherograms of RNA recovered from all three IL solutions contained the same distinct peaks and no additional peaks were observed.

The RIN, as well as the rRNA ratio, for plant RNA in 20% (w/v) choline glutamate was statistically similar to the RNA control, as demonstrated in Fig. 2. This indicates that 20% (w/v) choline glutamate provided near-perfect immediate stabilization, completely preventing any measurable degradation during the incubation period. Plant RNA recovered from 20% (w/v) choline dihydrogen phosphate was negligible and fell below the detection limit of the Qubit broad range assay. RNA was also not detected in the electrophoretic run. It was observed that the aqueous solution of choline dihydrogen phosphate was acidic and adjusting the pH of the aqueous choline dihydrogen phosphate solution to 7.0 with choline hydroxide facilitated RNA recovery upon alcohol precipitation. Buffered choline dihydrogen phosphate was previously identified as an efficient stabilization medium for siRNA and recombinant small RNA [19,21]. Although the RIN for plant RNA in 20% (w/v) buffered choline dihydrogen phosphate was statistically similar to that of the RNA control, the rRNA ratio was lower. The 20% (w/v) choline formate solution was found to be the least effective stabilizer for overall integrity, demonstrating a clear decrease in the RIN as well as the rRNA ratios. The low rRNA ratios for 20% (w/v) buffered choline dihydrogen phosphate and 20% (w/v) choline formate is attributed to the decrease in the height of the 25S rRNA peak.

RNA exhibits strong UV absorption at 260 nm. The purity of RNA was assessed using the ratios of UV absorbance at 260 and 280 nm (A260/280) and at 260 and 230 nm (A260/230). For pure RNA, the A260/280 ratio is typically close to 2.0 whereas the A260/230 ratio is 2.0–2.2 [33]. A noticeably lower A260/280 ratio suggests contamination by proteins or other substances that absorb strongly at or near 280 nm while a lower A260/230 indicates the presence of contaminants such as carbohydrates, ethylenediaminetetraacetic acid (EDTA), guanidine isothiocyanate, phenol or other substances that absorb at around 230 nm [34]. The RNA control and recovered RNA from the 20% (w/v) aqueous IL solutions demonstrated similar UV absorption spectra with the characteristic UV absorption band centered around 260 nm (Fig. 3) with the A260/280 ratio >1.80 , as observed in Table 1. The A260/230 ratio, which serves as a secondary measure of RNA purity, was also greater than 1.80. However, the RNA recovered from the 20% (w/v) choline glutamate solution

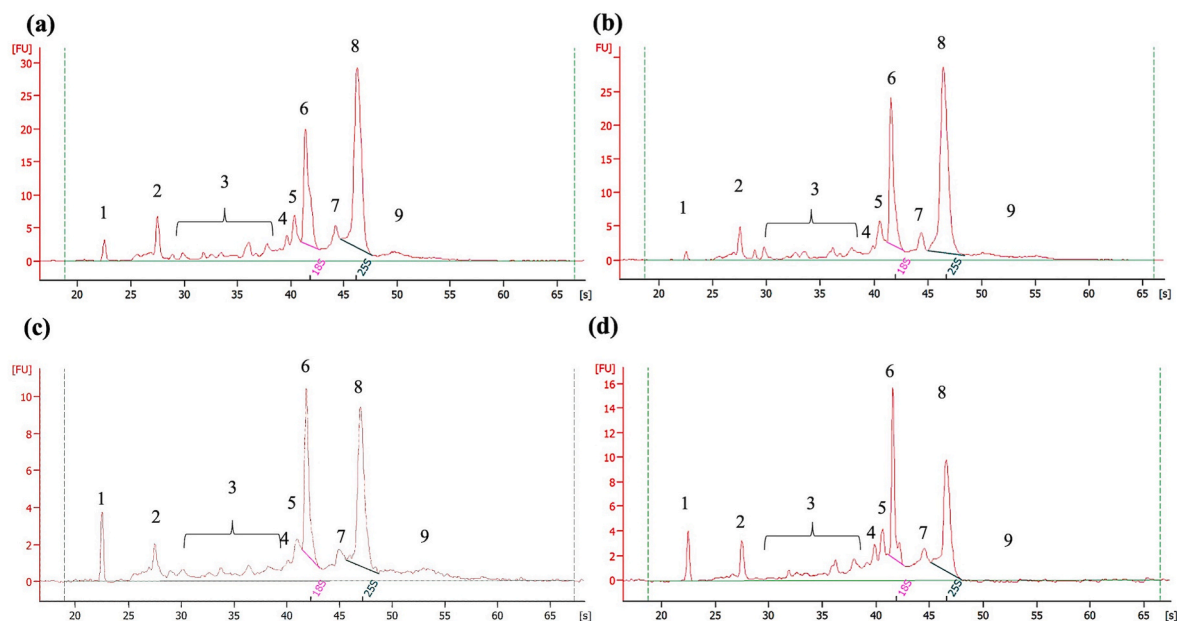


Fig. 1. Representative electropherograms for plant RNA after storing in IL mixtures for 0.5 h at room temperature (a) RNA control; RIN = 8.3, rRNA Ratio [25S/18S] = 2.3 (b) RNA recovered from 20% (w/v) choline glutamate; RIN = 8.2, rRNA Ratio [25S/18S] = 2.1 (c) RNA recovered from 20% (w/v) buffered choline dihydrogen phosphate; RIN = 7.4, rRNA Ratio [25S/18S] = 1.4. (d) RNA recovered from 20% (w/v) choline formate; RIN = 6.8, rRNA Ratio [25S/18S] = 1.2. Incubation time = 30 min; temperature = room temperature. 1 = lower marker (Internal size standard); 2 = small RNAs, 3 = fast RNAs; 4,5,7 = plastid rRNAs; 6 = 18S rRNA, 8 = 25S rRNA; 9 = post region.

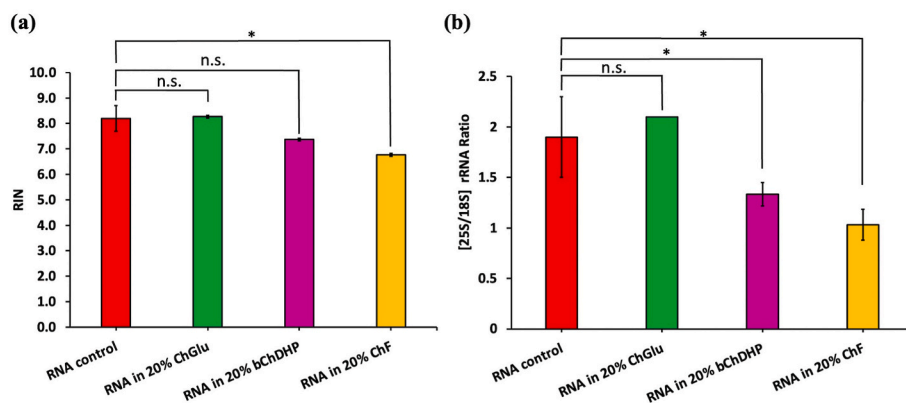


Fig. 2. Influence of different choline-based IL aqueous mixtures on plant RNA integrity in terms of (a) RIN and (b) 25S/18S rRNA ratio. Plant RNA (concentration = 150 ± 10 ng/ μ L) was incubated in 20% (w/v) IL aqueous mixtures at room temperature for 0.5 h and recovered by ethanol precipitation. Statistical significance was determined using the Student's t-test ($n = 3$). n.s. (no significance); * $p < 0.05$. Error bars represent the standard deviation. (ChGlu: choline glutamate; bChDHP: buffered choline dihydrogen phosphate; ChF: choline formate).

demonstrated a slightly elevated trough at 230 nm leading to a A260/230 ratio of 1.66 ± 0.02 . The RNA stock used for both the control and the aqueous IL solutions was the same; therefore, any deviation in purity ratios is likely attributed to the storage solution rather than the initial sample quality. Given that the RNA control exhibited ideal ratios, this slight decrease is not likely due to residual contaminants, such as carbohydrates or EDTA, but rather a minor increase in the absorbance of residual IL at 230 nm.

3.2. Temperature dependent degradation of plant RNA

Temperature is a critical factor influencing RNA integrity as high temperatures irreversibly damage RNA integrity [35]. To determine the protective capability of ILs against thermal degradation of RNA, it was incubated in 20% (w/v) IL solutions at different temperatures. Fig. 4 illustrates the temperature-dependent degradation of the plant RNA

control and the stabilizing effects of IL mixtures compared to an untreated control, in terms of RIN and rRNA ratio. Results of the temperature-dependent degradation studies for each of the treatments is provided in the following four paragraphs, for the RNA control, 20% (w/v) choline glutamate, 20% (w/v) buffered choline dihydrogen phosphate, and 20% (w/v) choline formate, respectively.

The RNA control exhibited intrinsic stability at different temperatures when stored without a specific stabilizer. The RIN values remained relatively high (>7.0) and were consistent from room temperature up to 60°C , as shown by Table S1. The RNA profiles remained highly intact within this temperature range, as indicated by sharp and well-resolved 18S and 25S rRNA peaks, as seen in Fig. S4(a)–(d), with average rRNA ratios at or above 1.5 (Fig. 4(b)). A critical drop in RNA integrity to 6.1 ± 0.1 , as well as the rRNA ratio to 1.0 ± 0.1 , was observed at 70°C . At 80°C , extensive degradation of RNA in the control was observed. The characteristic rRNA peaks were largely lost and replaced by a diffuse,

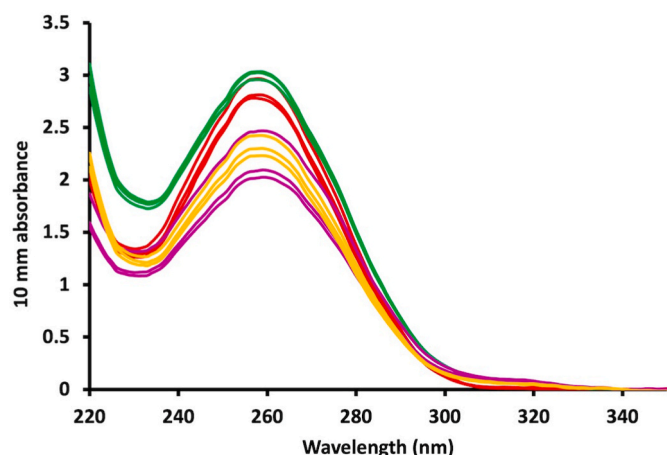


Fig. 3. NanoDrop UV absorption spectra for RNA control and RNA recovered after storing in IL mixtures for 0.5 h at room temperature. (Red: RNA control; Green: RNA in 20% (w/v) choline glutamate; Purple: RNA in 20% (w/v) buffered choline dihydrogen phosphate; Yellow: RNA in 20% (w/v) choline formate; $n = 3$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1

Nanodrop purity ratios of RNA after short-term exposure to aqueous solutions of choline-based ionic liquids at room temperature compared with water control. (ChGlu: choline glutamate; bChDHP: buffered choline dihydrogen phosphate; ChF: choline formate; $n = 3$).

Sample	Time (h)	A260/280	A260/230
RNA control	0.5	2.15 ± 0.03	2.18 ± 0.01
RNA in 20% (w/v) ChGlu	0.5	1.98 ± 0.01	1.66 ± 0.02
RNA in 20% (w/v) bChDHP	0.5	1.86 ± 0.01	1.87 ± 0.01
RNA in 20% (w/v) ChF	0.5	2.02 ± 0.01	1.87 ± 0.02

elevated baseline across the electropherogram as observed in Fig. S4(f). The RIN approached its lowest value at 90 °C indicative of complete RNA degradation. The electropherogram displayed a broad smear with no distinct rRNA peaks, as shown in Fig. S4(g), decreasing the RIN to 1.8 with a rRNA ratio of 0.0.

RNA in 20% (w/v) choline glutamate provided the best overall protection and maintained a significantly higher RIN compared to that of the control at room temperature and 70 °C and above (as shown in Table S2), demonstrating the best overall thermal stability. While a

statistically significant decrease in RIN was observed at temperatures above room temperature due to the high precision of the replicates, the RIN values were greater than 7.0 from room temperature up to 70 °C, as demonstrated in Fig. 4(a) and Table S1, and the 25S/18S rRNA ratios were ≥ 1.5 as shown in Table S3. The rRNA peaks remained sharp and well-resolved (See Fig. S5(a)–(e)). Even at 80 °C, identifiable sharp and well-resolved rRNA peaks were observed (See Fig. S5(f)), with a RIN value of 6.9 ± 0.1 and rRNA ratio of 1.5 ± 0.1 whereas complete collapse of the rRNA peaks was observed for the RNA control. The decline in RNA integrity in 20% (w/v) choline glutamate was slower, suggesting its protective mechanism remains largely functional even at 90 °C. At 90 °C, the RIN dropped to 5.5 ± 0.1 but remained much higher, compared to the RNA control (< 2.0). The 20% (w/v) choline glutamate solution provided the best protection to the rRNA structure even at temperatures of 80–90 °C where RNA control was fully degraded.

The 20% (w/v) buffered choline dihydrogen phosphate solution provided only limited thermal stabilization to RNA and was not as protective as 20% (w/v) choline glutamate. RNA degradation was found to be slower and less severe compared to that of the control and maintained higher RIN values at elevated temperatures (70 °C to 90 °C), as shown in Fig. 4(a). The RIN values from room temperature up to 70 °C were ≥ 7.0 as shown by Table S2. Despite the higher RIN values, the rRNA ratios from room temperature up to 70 °C were lower than the control (as shown by Table S4) due to the lower peak height of 25S rRNA compared to the 18S rRNA, as observed in Fig. S6(a)–(e). Beyond 70 °C, the RIN decreased to 6.2 ± 0.1 at 80 °C and 3.7 ± 0.1 at 90 °C. At 90 °C, the 25S rRNA peak completely disappeared (See Fig. S6(g)).

Among all IL compositions tested, RNA in 20% (w/v) choline formate exhibited the most severe thermal degradation and performed even worse than the control. The RIN values for this IL were consistently lower than the control at almost all temperatures (from 25 to 70 °C) and showed a rapid decline beginning at 50–60 °C, reaching values of ~ 2 at 70 °C indicative of extensive degradation. This trend was further supported by the 25S/18S rRNA ratio, which dropped sharply with increasing temperature and approached nearly zero by 60 °C, demonstrating near-complete loss of the 25S rRNA band as shown in Fig. S7(d). All rRNA peaks disappeared beginning from 70 °C to 90 °C, as seen in Fig. S7(e)–(g). Collectively, these results demonstrate that 20% (w/v) choline formate provided minimal protection against thermal degradation and may even accelerate RNA degradation, likely due to its destabilizing ionic environment. Therefore, choline formate was found to be the least effective IL evaluated in this study for RNA stabilization under elevated temperature conditions.

Among the different IL formulations tested, 20% (w/v) choline

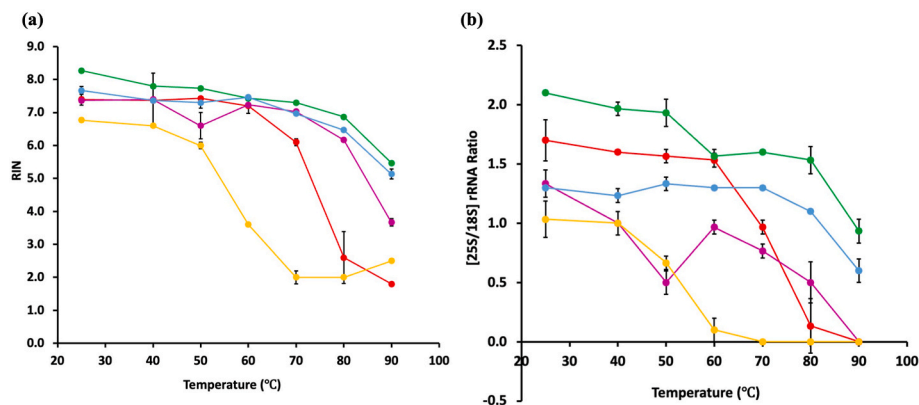


Fig. 4. The temperature-dependent degradation of plant RNA and the stabilizing effects of IL mixtures compared to the untreated control monitored using the (a) RIN and (b) 25S/18S rRNA ratio. Plant RNA (concentration = 150 ± 10 ng/ μ L) was incubated in 20% (w/v) IL aqueous mixtures at room temperature for 0.5 h at different temperatures ranging from room temperature and 90 °C and recovered after ethanol precipitation. (Red: RNA control; Green: RNA in 20% (w/v) choline glutamate; Purple: RNA in 20% (w/v) buffered choline dihydrogen phosphate; Yellow: RNA in 20% (w/v) choline formate; Blue: RNA in 50% (w/v) choline glutamate; $n = 3$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

glutamate provided near-perfect immediate stabilization at room temperature and RIN values > 7.0 and 25S/18S rRNA ratios > 1.5 for the majority of the temperatures tested. Therefore, this IL formulation was established as a high performing baseline for plant RNA stabilization. To investigate whether increasing the IL concentration, thereby increasing the ionic strength and potential protective effect would offer synergistic thermal protection, subsequent experiments were conducted with 50% (w/v) choline glutamate as a high concentration benchmark. As shown in Fig. 4(a) and Table S2, the RIN values for RNA in 50% (w/v) choline glutamate were comparable to that of the control up to 60 °C and higher than the control from 70 °C to 90 °C ($p < 0.0001$), indicative of its protection against thermal degradation of RNA at higher temperatures. Despite its higher concentration, 50% (w/v) choline glutamate failed to match the thermal stability offered by the 20% (w/v) IL. This trend was further corroborated by the 25S/18S rRNA ratios in Fig. 4(b) and Table S4, where RNA in 50% (w/v) choline glutamate had a lower ratio than the 20% (w/v) choline glutamate across the entire temperature range, supporting that 20% (w/v) choline glutamate was the optimal concentration for maintaining higher RNA integrity across a wide temperature range without the added viscosity and potential purity interference (as seen in A260/A230 shifts in Table S5) associated with higher IL concentrations. A comparison of the Bioanalyzer gel profiles for the RNA control and RNA in different IL formulations is provided in Fig. S8.

Nanodrop UV spectrophotometric analysis was employed to assess the purity of RNA samples following incubation for 0.5 h across the temperature range from room temperature up to 90 °C. Despite the decline in integrity, the RNA control maintained high A260/280 and A260/230 ratios across all tested temperatures, as shown Table S5, indicating that it is possible for RNA to be pure (i.e., have ideal A260/280 and A260/230 ratios) yet be degraded. Degradation often means the RNA molecule has been fragmented into smaller pieces, but these fragments still maintain their inherent UV absorption characteristics, leading to seemingly good purity ratios despite loss of integrity. In contrast, RNA recovered from 20% (w/v) aqueous IL solutions led to a noticeable decrease in the A260/230 purity ratio, likely due to residual IL absorbing at 230 nm. The A260/280 ratios remained closer to the acceptable range of approximately 2.0 across most samples.

3.3. Effect of ILs on RNA conformation and melting temperature

CD spectroscopy was used to measure any changes to RNA conformation in the presence of ILs. The CD spectra of the studied plant RNA control sample (Fig. S9(a)) and RNA in 20% (w/v) choline dihydrogen phosphate (Fig. S9(b)) demonstrated A-type structure with a positive Cotton effect near 265 nm and a negative band near 210 nm [36]. The CD spectra of plant RNA in 20% (w/v) choline glutamate and 20% (w/v) choline formate had high interference from the IL solution at lower wavelengths, as seen in Fig. S9(c) and (d). The high tension (HT) voltage value of the machine Jasco spectropolarimeter crossed beyond 700 at approximately 227 nm and beyond this HT voltage value, the CD signal became disproportionate. It has been reported that ILs have high interference at lower wavelengths [21,22]. Therefore, information on the secondary structure of RNA below 230 nm could not be gained for the 20% (w/v) choline glutamate and 20% (w/v) choline formate IL solutions. On the other hand, RNA in all three IL solutions demonstrated a positive Cotton effect around 265 nm.

RNA melting experiments were additionally performed at 265 nm to evaluate the stability of the secondary structure of RNA in the presence of ILs, assessed by the RNA melting temperature (T_m). The T_m of RNA dissolved in RNase-free water was 56 °C, as shown in Fig. 5. In the presence of 20% (w/v) choline glutamate, the T_m of plant RNA increased by 4 °C whereas for the 20% (w/v) choline dihydrogen phosphate solution it decreased by 3 °C and for 20% (w/v) choline formate by 9 °C. The increase in T_m in the presence of 20% (w/v) choline glutamate suggests that it thermodynamically stabilizes the RNA secondary structure. Similar findings were reported by Tulsian et al. who observed an increment of T_m by 5 °C for RNA sourced from *Torula* yeast in the presence of 50 mM choline amino acid ILs [37]. The structural reinforcement offered by 20% (w/v) choline glutamate, combined with its ability to hinder degradation, lead to the observed best overall thermal stability. The drastic drop in T_m in the presence of 20% (w/v) choline formate indicates that the RNA structure is significantly destabilized. When the structure is weakened, it becomes highly susceptible to degradation leading to complete collapse at low temperatures which was observed earlier with the low RIN values. This

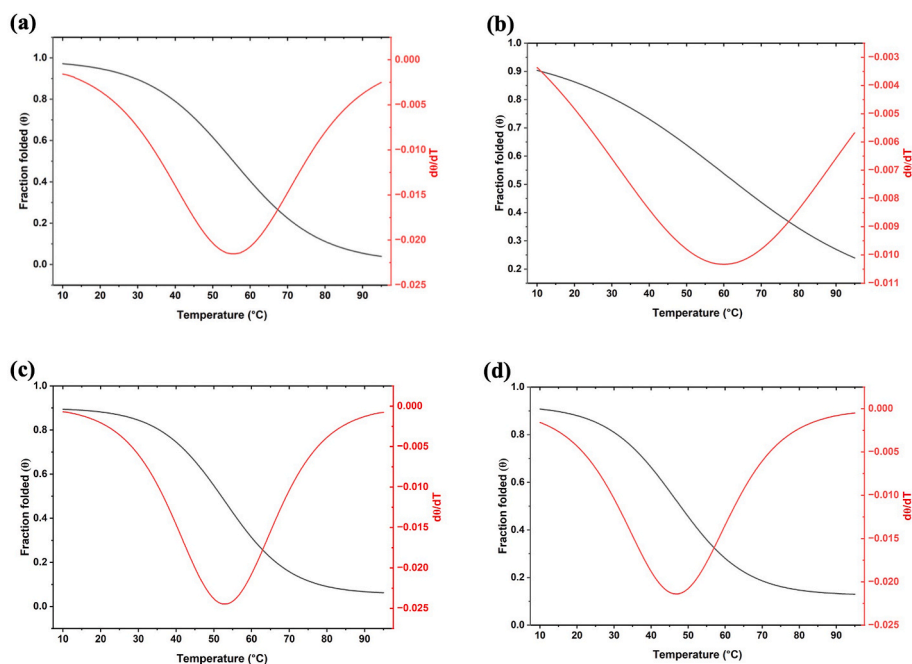


Fig. 5. Fraction folded curves (black) along with their first derivatives (red) as a function of temperature for plant RNA incubated for 1 h at room temperature in (a) water, (b) 20% (w/v) choline glutamate, (c) 20% (w/v) buffered choline dihydrogen phosphate and (d) 20% (w/v) choline formate. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

confirms that a severely destabilized structure accelerates degradation.

3.4. RNA stability at elevated temperature for extended periods of time

Given the high thermal stability of RNA in 20% (w/v) choline glutamate, the long-term thermal stability of RNA was investigated by incubating RNA at 40 °C for 72 h. This temperature was selected as a representative temperature to simulate the elevated temperature conditions in summer during which the temperature rises above 25 °C. As shown in Fig. 6, the RNA control degraded slowly up to 12 h, but the degradation became more prominent between 24 h and 72 h. The RIN values decreased from 7.3 ± 0.1 to 2.9 ± 0.1 by 72 h, and the 25S/18S ratio declined from approximately 1.6 to nearly zero as shown in Tables S6 and S8 respectively. The 25S rRNA peak gradually decreased compared to the 18S rRNA peak over time and almost completely disappeared after 72 h, as seen in Fig. S10. This loss of integrity is consistent with the tendency of RNA to degrade at elevated temperatures [35]. These results confirm that RNA stored in water undergoes significant degradation when exposed to heat for extended periods of time.

In contrast, RNA incubated in 20% (w/v) choline glutamate demonstrated distinctly improved thermal stability. The RIN values for RNA in 20% (w/v) choline glutamate decreased slowly and maintained values nearly 6.0 and higher than the control throughout the 72 h incubation period as shown in Table S7. Similarly, the 25S/18S rRNA ratio remained greater than 1.0 (as shown in Table S8) over the entire time course, indicative of less degradation for the 25S rRNA compared to that of the control. Even after 72 h, the rRNA peaks were distinct and prominent although the peak height of 25S rRNA was slightly lower compared to that of the 18S rRNA, as demonstrated in Fig. S11. A comparison of the Bioanalyzer gel profiles of the RNA control and recovered RNA in 20% (w/v) choline glutamate incubated at 40 °C is shown in Fig. S12. The reduced degradation rate for RNA in 20% (w/v) choline glutamate suggests that the IL environment lessens thermally-driven degradation, extending the lifetime of RNA at elevated temperatures and supporting its potential application as a promising medium for thermally sensitive RNA workflows or as a stabilizing medium for intermediate storage or transportation of isolated RNA in the absence of temperature control.

It has been established that the stability of RNA can be affected by the initial concentration [23]. To validate the robustness of the IL platform across varying initial RNA loads, additional stability studies were conducted at 2 lower concentrations of 75 ng/ μ L and 15 ng/ μ L. As shown in Fig. S13, the aqueous RNA control at 75 ng/ μ L exhibited higher sensitivity to degradation compared to the 150 ng/ μ L control at 24 and 48 h. In contrast, the integrity of RNA stabilized by the IL remained statistically similar ($p > 0.05$) across both concentrations in terms of RIN and 25 S/18S rRNA ratios. Furthermore, the

electropherograms in Fig. S14 indicate that even at a 10-fold dilution (15 ng/ μ L), where RNA is highly susceptible to degradation, the IL platform successfully preserved the 18S and 25S rRNA subunits. While the 15 ng/ μ L concentration precluded a reliable software generated RIN for RNA recovered by the IL, the electropherograms clearly demonstrate intact peaks compared to the fully degraded aqueous controls. This indicates that the stabilization efficacy remains effective across a wide concentration range, including lower concentrations making the method suitable for yield-limited RNA samples.

3.5. Time-dependent RNA stability at room temperature

Given the observed thermal stabilization, the long-term efficacy of 20% (w/v) choline glutamate as an RNA preservation medium at room temperature was investigated for up to ten weeks. Fig. 7 shows a continuous loss in integrity over time, in terms RIN and rRNA ratio, for the RNA control. By the end of 6 weeks, the RIN values dropped from 7.3 ± 0.1 to 1.8 and rRNA ratios from 1.3 to 0.0, indicating complete degradation and loss of RNA integrity. This degradation was observed in the electropherograms of the RNA control in Fig.S15(a)-(h), which transitioned from intact 18S and 25S rRNA peaks to a broad, smeared pattern. At the end of 7 weeks, although the RIN value increased compared to that in 6 weeks, the electropherogram for RNA recovered after 7 weeks in Fig. S15(h) demonstrated a higher shift in the RNA size distribution towards smaller fragments, indicative of severely degraded RNA. During the 8th week, no RNA was recovered from the control.

The 20% (w/v) choline glutamate solution demonstrated superior RNA stability by delaying degradation by several weeks compared to that of the control. While the RNA control was completely degraded by 6 weeks, Fig. 7 shows that RIN values and rRNA ratios for the IL remained at 5.2 ± 0.1 and 0.6 ± 0.1 , respectively, reflecting its protective effect. While the electropherograms for RNA in 20% (w/v) choline glutamate showed a decline in height for the 25S rRNA peak, both rRNA peaks were visible even until the end of 10 weeks as shown in Fig. S16(a)-(k). These data indicate that 20% (w/v) choline glutamate effectively slows RNA degradation at ambient temperatures, providing prolonged stabilization compared to RNase-free water. A comparison of the Bioanalyzer gel profiles for the weekly stability of RNA control and RNA recovered from 20% (w/v) glutamate is shown in Fig. S17.

Despite the progressive decline in integrity, Table 2 shows that the RNA control maintained high A260/280 and A260/230 ratios falling within the expected range for pure RNA, indicative of an absence of contaminants that absorb at 230 nm and 280 nm. However, RNA recovered from 20% (w/v) choline glutamate exhibited slightly reduced ratios for both metrics. While the A260/280 ratios were approximately 2.0, the average A260/230 ratios ranged from 1.04 to 1.67. The lower A260/230 ratios were likely due to residual IL that absorbs at 230 nm.

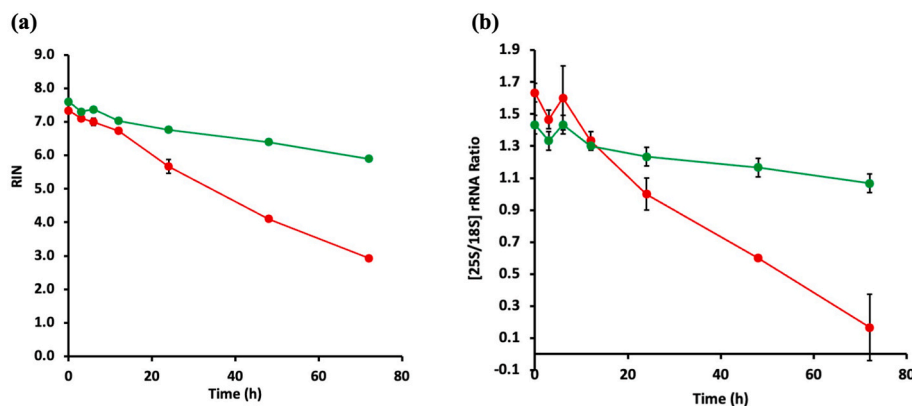


Fig. 6. Thermal stability of plant RNA at 40 °C for an extended time period in terms of (a) RIN and (b) 25S/18S rRNA ratio. (Red: RNA control; Green: RNA in 20% (w/v) choline glutamate; $n = 3$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

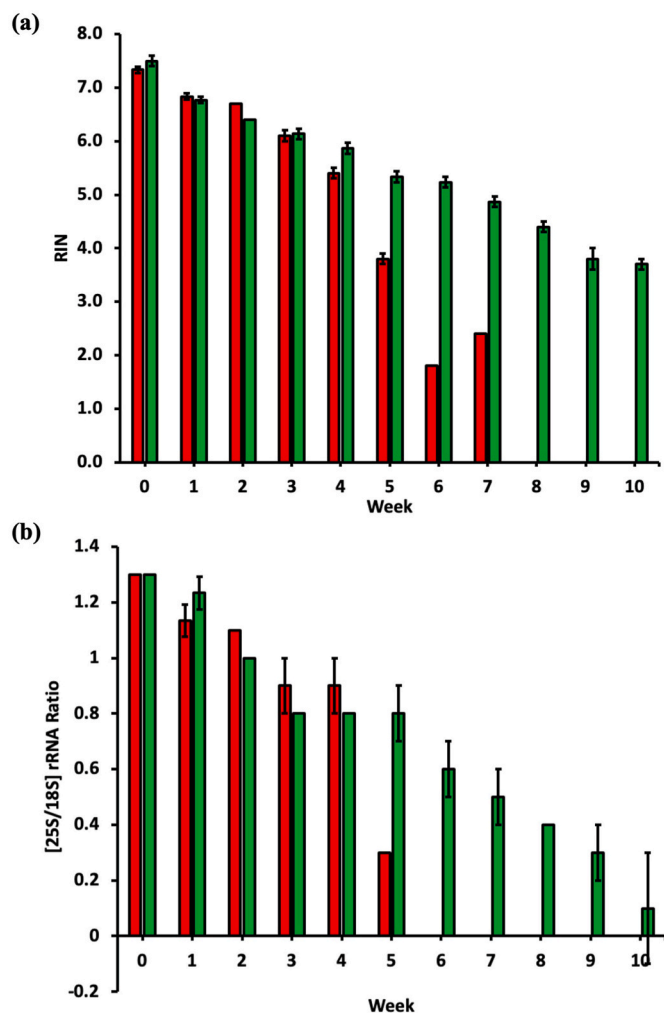


Fig. 7. Long-term stability of plant RNA stored in water (red) and 20% (w/v) choline glutamate (green) as measured by (a) RIN and (b) 25S/18S rRNA ratio. ($n = 3$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

Nanodrop purity ratios of RNA after incubation in water and 20% (w/v) choline glutamate (ChGlu) at room temperature. ($n = 3$).

Sample	Time (Weeks)	A260/280	A260/230
RNA control	0	2.14 ± 0.00	2.05 ± 0.02
	1	2.15 ± 0.01	1.99 ± 0.04
	2	2.17 ± 0.01	2.02 ± 0.09
	3	2.01 ± 0.20	1.61 ± 0.61
	4	2.12 ± 0.01	1.96 ± 0.04
	5	2.12 ± 0.01	2.02 ± 0.04
	6	2.13 ± 0.01	2.04 ± 0.02
RNA in 20% (w/v) ChGlu	7	2.15 ± 0.00	2.04 ± 0.01
	0	1.94 ± 0.05	1.30 ± 0.11
	1	1.93 ± 0.04	1.39 ± 0.09
	2	1.98 ± 0.05	1.62 ± 0.09
	3	1.96 ± 0.01	1.56 ± 0.08
	4	1.98 ± 0.06	1.51 ± 0.10
	5	1.86 ± 0.09	1.04 ± 0.13
	6	1.85 ± 0.02	1.27 ± 0.30
	7	2.03 ± 0.03	1.67 ± 0.13
	8	1.99 ± 0.03	1.52 ± 0.02
	9	1.98 ± 0.01	1.48 ± 0.01
	10	2.00 ± 0.02	1.41 ± 0.23

Importantly, the absorption maximum was centered around 260 nm without any shift over the entire incubation period for both conditions, as seen in Fig. S18.

3.6. Successive recycling and reuse of IL for plant RNA preservation

A key objective of this study was to establish a sustainable RNA preservation process. Following the recovery of plant RNA, the feasibility of recycling and reuse of IL for plant RNA storage was evaluated. As illustrated in Fig. 8, RNA was recovered via ethanol-induced precipitation, and the choline glutamate IL was recovered from the supernatant by removing the solvent using rotary evaporation until a constant weight was achieved. The concentration of recovered IL was adjusted with water to 20% (w/v) and reused for a new batch of plant RNA. This cycle was successfully repeated for 5 consecutive iterations. The structural integrity of the recycled IL at the end of 5th cycle was found to be the same as the pure IL, as observed in Fig. S19. The amount of IL recovered at the end of 1st and 2nd cycle was 91–92.5% (w/w), with the 3rd–5th cycle approaching 75–77% (w/w). RNA regenerated from recycled choline glutamate had similar purity and integrity with no degradation compared to those of the RNA control for all 5 cycles, as shown in Table S10 and Fig. 9, respectively. These results were further supported by Bioanalyzer gel profiles in Fig. S20. By precipitating RNA from a bulk solution and reconstituting it in a smaller volume, Table S10 reveals that RNA preconcentration was achieved thereby improving the sensitivity for downstream analysis. The likely interference of the IL with absorbance at 230 nm which was observed previously was prominent for RNA that was recovered from 10 μ L of aqueous IL solutions. In contrast, this interference was not observed as shown by the ideal A260/A230 values in Table S10 for RNA recovered from bulk solutions (150–250 μ L), where RNA preconcentration was achieved. This indicates that when higher concentration of RNA is recovered, the interference of the residual IL becomes negligible.

Although direct spectroscopic or computational evidence of the IL-RNA interaction was not collected in this study, the observed experimental trends allow us to gain insights on the stabilization mechanism. The specific chemical structure of both the cation and anion within an IL influences its ability to stabilize plant RNA, allowing for the design of ILs with optimized performance for preservation. Choline-based ILs are particularly significant due to their biocompatibility, low toxicity, and biodegradability, making them suitable for complex biological applications. It has been reported that choline cation has a strong site

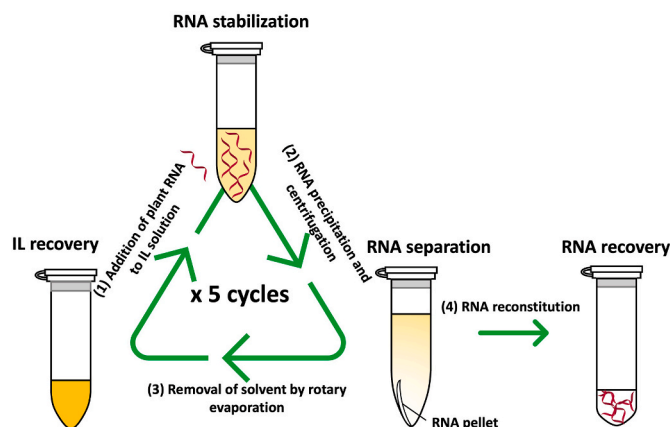


Fig. 8. Schematic diagram for recyclable IL-based workflow for plant RNA stabilization, IL recovery, and reuse. (1) Plant RNA is dissolved in 20% (w/v) choline glutamate IL solution (2) RNA is selectively precipitated from the IL solution by ethanol followed by centrifugation (3) The supernatant is separated and the IL is regenerated by solvent removal using rotary evaporation and reused for subsequent RNA stabilization cycles (4) The precipitated RNA is recovered by reconstitution in RNase-free water for downstream analysis.

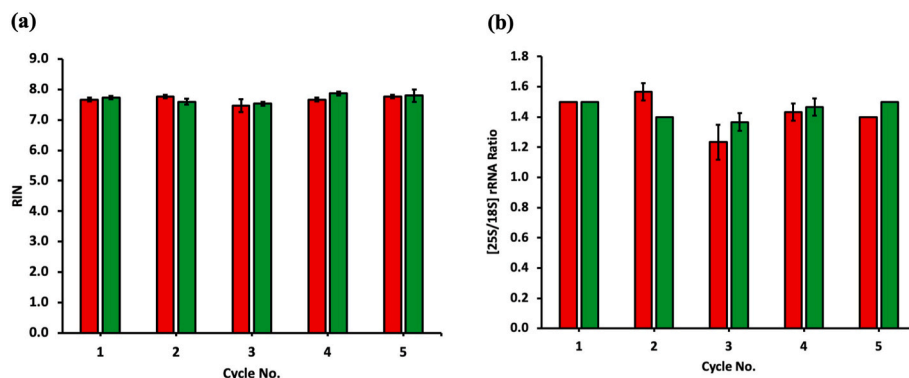


Fig. 9. Effect of five consecutive IL recycling cycles on plant RNA integrity in terms of (a) RIN and (b) 25S/18S rRNA ratio. (Red: RNA control; Green: RNA recovered from 20% (w/v) choline glutamate; $n = 3$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

preference for the minor groove of the RNA structure and RNA-IL binding is driven by electrostatic interaction [37]. The anionic counterpart of the IL is shown to exert a significant influence on thermodynamic stability. The increase in the T_m of plant RNA in 20% (w/v) choline glutamate aligns with studies indicating that this IL reinforces the secondary structure of RNA and enhance thermal stability [22,37]. Although amino acid anions such as glutamate contribute minimally to direct binding, they are essential for charge stabilization and structural reinforcement [37]. This is evidenced by the contrast in performance compared to choline formate and choline dihydrogen phosphate ILs. The decrease in T_m , RIN values, and rRNA ratios in 20% (w/v) choline formate suggests a destabilizing ionic environment. Furthermore, non-buffered choline dihydrogen phosphate solutions are highly acidic and can promote RNA hydrolysis and adjusting the pH to neutral with choline hydroxide was necessary to facilitate RNA recovery and stability. A range of sophisticated biophysical and computational techniques need to be employed to elucidate the intricate interactions between ILs and plant RNA, providing detailed insights into stabilization mechanisms.

To evaluate the practical utility of the IL platform for RNA stabilization, its performance and logistical requirements were compared against two commercial standards: GenTegra-RNA and RNAsheer (see Table S11 in the Supporting Information). While these commercial platforms have demonstrated exceptional stability for RNA derived primarily from human and animal sources (e.g., HeLa cells, rat liver, pancreas, etc.), the focus of the present study was to optimize a platform for RNA derived from plants which often presents unique challenges due to high concentrations of secondary metabolites and RNases. Unlike RNAsheer, which necessitates specialized encapsulation hardware and GenTegra-RNA which requires complete sample drying during dry storage, the IL system provides RNA stabilization in the liquid phase. This significantly simplifies field-to-lab workflows by eliminating the need for drying or vacuum sealing in remote collection sites. Furthermore, the IL platform distinguishes itself through its environmental and economic profile. Unlike single-use stabilization kits, this IL is recyclable for at least five cycles and is composed of non-toxic, biodegradable components. With a significantly lower cost per sample, this method offers a sustainable alternative for high-throughput plant research where budget constraints often limit large scale genomic studies. Large scale evaluations of plant transcriptome sequencing have demonstrated that maximizing the RIN and A260/A230 ratios will improve the size of assembled transcriptomes [38]. The primary advantage of total plant RNA stabilization platform employing 20% (w/v) choline glutamate lies in maintaining higher RNA integrity during storage and handling, under ambient and particularly, elevated temperature conditions where degradation is a major challenge. The intended application for this IL-based platform is to facilitate the storage and transport of plant RNA

at ambient temperature and elevated temperature conditions without the need for dry ice during transit. The ability to reuse choline glutamate without compromising RNA integrity or purity represents a significant advancement in green analytical chemistry. By replacing traditional, single-use stabilization reagents with a biocompatible and recyclable IL, this method significantly reduces chemical waste by establishing a closed-loop system for the solvent. The use of a biocompatible IL, in a recyclable workflow, minimizes the environmental footprint and reagent costs associated with large-scale plant genetic resource preservation.

4. Conclusions

This study established that the 20% (w/v) choline glutamate solution is a highly effective, biocompatible, and recyclable RNA stabilization medium capable of preserving total plant RNA. Among the IL formulations evaluated, 20% (w/v) choline glutamate consistently provided the most robust RNA stabilization across all tested conditions, including short-term handling, elevated temperatures, long-term storage, and repeated solvent recycling. Immediate stabilization experiments demonstrated that 20% (w/v) choline glutamate preserved RNA integrity at levels statistically similar to that of RNA controls, as confirmed by both RIN values and rRNA ratios. Thermal stability experiments further revealed the exceptional protective ability of 20% (w/v) choline glutamate for RNA. While control RNA degraded at temperatures from 70 °C and above, RNA stored in 20% (w/v) choline glutamate retained higher RIN values and distinct rRNA peaks even at 80–90 °C, indicating a substantial decrease in thermal degradation. These observations were corroborated by CD melting experiments, which showed an increase in RNA melting temperature in choline glutamate, confirming thermodynamic stabilization of RNA secondary structure.

Long-term storage studies demonstrated higher integrity for plant RNA in 20% (w/v) choline glutamate incubated at 40 °C for at least 72 h, whereas control RNA underwent rapid degradation. At room temperature, 20% (w/v) choline glutamate delayed complete RNA degradation by several weeks. Importantly, UV spectrophotometric analyses demonstrated that RNA purity metrics alone are insufficient indicators of RNA quality, as degraded RNA can still exhibit ideal A260/280 and A260/230 ratios. This emphasizes the necessity of integrity-based metrics, such as RIN and electrophoretic profiles, when evaluating RNA preservation strategies. The successful recycling and reuse of choline glutamate for at least five consecutive cycles without loss of RNA integrity or IL structure established this approach as a sustainable RNA preservation alternative, significantly reducing chemical waste and reagent consumption.

The IL-based platform described in this study offers a promising foundation for the development of standardized, green RNA

preservation workflows with broad applicability in plant molecular studies. While this study establishes the fundamental stability and recyclability of choline glutamate for plant RNA, future studies should focus on the compatibility of IL-preserved RNA with RT-qPCR and library preparation protocols for Next-Generation Sequencing. The feasibility of such applications is strongly supported by previous reports, which demonstrated that plant DNA extracted and stabilized by IL formulations is highly suitable for PCR and whole-genome Illumina sequencing yielding >99% whole genome coverage [28]. Given these established successes in DNA genomics, exploring RNA stabilized by ILs for transcriptomic scale assays is a direct and promising possibility. Evaluating the quality of RNA for adapter ligation or reverse transcription at a genomic scale will be critical for large scale agricultural transcriptomics. Furthermore, practical, real world evaluation of IL-based preservation media for isolated RNA should be encouraged not only for storage and use of preserved RNA, but also for transporting isolated RNA from lab to lab, or from the field to the laboratory. These studies should aim to test IL-preserved RNA from a larger diversity of taxa across the plant tree of life as well as a larger diversity of tissues. The unstable nature of isolated RNA has prevented its widespread use in many research applications; limits its practical use in many fields and makes it prohibitively expensive for many applied and basic research laboratories. An affordable, safe, recyclable solution for improving the preservation and storage of RNA would facilitate numerous applications of this nucleic acid for research and widescale applications as well.

CRediT authorship contribution statement

Shashini De Silva: Conceptualization, Investigation, Methodology, Validation, Visualization, Writing – original draft. **Cecilia Cagliero:** Methodology, Writing – review & editing. **Morgan R. Gestel:** Methodology, Writing – review & editing. **Gabriel Johnson:** Methodology, Writing – review & editing. **Jared L. Anderson:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.aca.2026.345501>.

Data availability

Data will be made available on request.

References

- [1] M.G. Caprara, T.W. Nilsen, RNA: versatility in form and function, *Nat. Struct. Biol.* 7 (2000) 831–833, <https://doi.org/10.1038/82816>.
- [2] L. Baronti, H. Karlsson, M. Marušić, K. Petzold, A guide to large-scale RNA sample preparation, *Anal. Bioanal. Chem.* 410 (2018) 3239–3252, <https://doi.org/10.1007/s00216-018-0943-8>.
- [3] S.L. Fordyce, M.-L. Kampmann, N.L. Van Doorn, M.T.P. Gilbert, Long-term RNA persistence in postmortem contexts, *Invest. Genet.* 4 (2013) 7, <http://www.investigativegenetics.com/content/4/1/7>.
- [4] Y.K. Kim, RNA therapy: rich history, various applications and unlimited future prospects, *Exp. Mol. Med.* 54 (2022) 455–465, <https://doi.org/10.1038/s12276-022-00757-5>.
- [5] L. Fang, L. Xiao, Y.W. Jun, Y. Onishi, E.T. Kool, Reversible 2'-OH acylation enhances RNA stability, *Nat. Chem.* 15 (2023) 1296–1305, <https://doi.org/10.1038/s41557-023-01246-6>.
- [6] J. Haensler, L. Even, P. Wils, F. Bensaid, A. Dias, H. Deng, S. Karve, F. DeRosa, Not so cold! improving the thermostability of mRNA vaccines, *Expert Rev. Vaccines* 24 (2025) 1149–1162, <https://doi.org/10.1080/14760584.2025.2596674>.
- [7] A.L. Fabre, M. Colotte, A. Luis, S. Tuffet, J. Bonnet, An efficient method for long-term room temperature storage of RNA, *Eur. J. Hum. Genet.* 22 (2014) 379–385, <https://doi.org/10.1038/ejhg.2013.145>.
- [8] M. Scott, M. Duckett, J. Modiano, C. Yang, H. Martinez, B. Iverson, R. Nunez, Ambient temperature stabilization of Feline and Canine tumor cell RNA for use in gene expression assays, *GenTegra Applic. Note* (n.d.), <https://gentegra.com/wp-content/uploads/2020/05/Apps-note-GenTegra-RNA-20140108.pdf> (accessed January 30, 2026).
- [9] GenTegra LLC, GenTegraRNA GTR5100-S and GTR5025-S user guide. <https://gentegra.com/wp-content/uploads/2025/11/GenTegraRNA-User-Guide-v3.2025.pdf>, 2025. (Accessed 17 January 2026).
- [10] E. Seelenfreund, W.A. Robinson, C.M. Amato, A.C. Tan, J. Kim, S.E. Robinson, Long term storage of dry versus frozen RNA for next generation molecular studies, *PLoS One* 9 (2014) e111827, <https://doi.org/10.1371/JOURNAL.PONE.0111827>.
- [11] Encapsulation of purified RNA, Imagene (n.d.), <https://imagene.eu/products-services-en/services-en/encapsulation-of-purified-rna/> (accessed January 17, 2026).
- [12] S. De Silva, I. Ocaña-Rios, C. Cagliero, M.R. Gestel, G. Johnson, J.L. Anderson, Isolation of DNA from plant tissues using a miniaturized matrix solid-phase dispersion approach featuring ionic liquid and magnetic ionic liquid solvents, *Anal. Chim. Acta* 1245 (2023) 340858, <https://doi.org/10.1016/J.ACA.2023.340858>.
- [13] S. De Silva, C. Cagliero, M.R. Gestel, G. Johnson, J.L. Anderson, Versatile DNA extraction from diverse plant taxa using ionic liquids and magnetic ionic liquids: a methodological breakthrough for enhanced sample utility, *Plant Methods* 20 (2024), <https://doi.org/10.1186/s13007-024-01217-z>.
- [14] K.K. Athira, R.L. Gardas, Anion-driven influence of tetrabutylammonium-based ionic liquids on DNA stability and interaction mechanisms, *ACS Phys. Chem. Au* 5 (2025) 387–397, <https://doi.org/10.1021/acspchemau.5c00015>.
- [15] R. Vijayaraghavan, A. Izgorodin, V. Ganesh, M. Surianarayanan, D.R. MacFarlane, Long-term structural and chemical stability of DNA in hydrated ionic liquids, *Angew. Chem. Int. Ed.* 49 (2010) 1631–1633, <https://doi.org/10.1002/anie.200906610>.
- [16] R.R. Mazid, A. Cooper, Y. Zhang, R. Vijayaraghavan, D.R. MacFarlane, C. Cortez-Jugo, W. Cheng, Enhanced enzymatic degradation resistance of plasmid DNA in ionic liquids, *RSC Adv.* 5 (2015) 43839–43844, <https://doi.org/10.1039/c5ra05518k>.
- [17] A.A. Quintana, A.M. Sztapka, V. de C. Santos Ebinuma, C. Agatemor, Enabling sustainable chemistry with ionic liquids and deep eutectic solvents: a fad or the future? *Angew. Chem.* 134 (2022) <https://doi.org/10.1002/ANGE.202205609>.
- [18] N.M. Abbasi, S. De Silva, A. Biswas, J.L. Anderson, Ultra-low viscosity and high magnetic susceptibility magnetic ionic liquids featuring functionalized diglycolic acid ester rare-earth and transition metal chelates, *ACS Omega* 8 (2023) 27751–27760, <https://doi.org/10.1021/ACSOMEGA.3C03938>.
- [19] R.R. Mazid, U. Divisekera, W. Yang, V. Ranganathan, D.R. MacFarlane, C. Cortez-Jugo, W. Cheng, Biological stability and activity of siRNA in ionic liquids, *Chem. Commun.* 50 (2014) 13457–13460, <https://doi.org/10.1039/c4cc05086j>.
- [20] B. Kang, S.V. Park, S.S. Oh, Ionic liquid-caged nucleic acids enable active folding-based molecular recognition with hydrolysis resistance, *Nucleic Acids Res.* 52 (2024) 73–86, <https://doi.org/10.1093/nar/gkad1093>.
- [21] A.Q. Pedro, P. Pereira, M.J. Quental, A.P. Carvalho, S.M. Santos, J.A. Queiroz, F. Sousa, M.G. Freire, Cholinium-based good's buffers ionic liquids as remarkable stabilizers and recyclable preservation media for recombinant small RNAs, *ACS Sustain. Chem. Eng.* 6 (2018) 16645–16656, <https://doi.org/10.1021/acssuschemeng.8b03900>.
- [22] M.V. Quental, A.Q. Pedro, P. Pereira, M. Sharma, J.A. Queiroz, J.A.P. Coutinho, F. Sousa, M.G. Freire, Integrated extraction-preservation strategies for RNA using biobased ionic liquids, *ACS Sustain. Chem. Eng.* 7 (2019) 9439–9448, <https://doi.org/10.1021/acssuschemeng.9b00688>.
- [23] U. Chheda, S. Pradeepan, E. Esposito, S. Strezsak, O. Fernandez-Delgado, J. Kranz, Factors affecting stability of RNA – temperature, length, concentration, pH, and buffering species, *J. Pharm. Sci.* 113 (2024) 377–385, <https://doi.org/10.1016/j.xphs.2023.11.023>.
- [24] H.M. Tetreault, Z. Zingerman, L. Hill, S. Ibrahim, J. Maschinski, K.D. Heineman, C. Walters, Assessing the RNA integrity in dry seeds collected from diverse endangered species native to the USA, *Front. Plant Sci.* 16 (2025) 1585631, <https://doi.org/10.3389/fpls.2025.1585631>.
- [25] Comparison of Fluorescence-based Quantitation with UV Absorbance Measurements, Thermo Fisher Scientific (n.d.), <https://documents.thermofisher.com/TFS-Assets/LSG/Technical-Notes/fluorescence-UV-quantitation-comparison-tech-note.pdf> (accessed January 13, 2026).
- [26] O. Mueller, K. Hahnenberger, M. Dittmann, Y. Herman, R. Dubrow, R. Nagle, D. Ilsley, A microfluidic system for high-speed reproducible DNA sizing and quantitation, *Electrophoresis* 21 (2000) 128–134, [https://doi.org/10.1002/\(SICI\)1522-2683\(20000101\)21:1<128::AID-ELPS128>3.0.CO;2-M](https://doi.org/10.1002/(SICI)1522-2683(20000101)21:1<128::AID-ELPS128>3.0.CO;2-M).

- [27] A. Schroeder, O. Mueller, S. Stocker, R. Salowsky, M. Leiber, M. Gassmann, S. Lightfoot, W. Menzel, M. Granzow, T. Ragg, The RIN: an RNA integrity number for assigning integrity values to RNA measurements, *BMC Mol. Biol.* 7 (2006), <https://doi.org/10.1186/1471-2199-7-3>.
- [28] S. De Silva, P.C. Bentz, C. Cagliero, M.R. Gostel, G. Johnson, J.L. Anderson, Ionic liquid-assisted seed genomic DNA extraction for advanced sequencing applications, *Plant Methods* 21 (2025), <https://doi.org/10.1186/S13007-025-01417-1>.
- [29] K. Fujita, D.R. MacFarlane, K. Noguchi, H. Ohno, Choline dihydrogen phosphate, *Acta Crystallogr. Sect. E Struct. Rep. Online* 65 (2009), <https://doi.org/10.1107/S1600536809007259>.
- [30] S. Fleige, M.W. Pfaffl, RNA integrity and the effect on the real-time qRT-PCR performance, *Mol. Aspect. Med.* 27 (2006) 126–139, <https://doi.org/10.1016/j.mam.2005.12.003>.
- [31] O. Mueller, S. Lightfoot, A. Schroeder, RNA integrity number (RIN)-standardization of RNA quality control application, *Agilent Technol. Application Note* (2016) 5989–591165EN.
- [32] M.B. Fleming, C.M. Richards, C. Walters, Decline in RNA integrity of dry-stored soybean seeds correlates with loss of germination potential, *J. Exp. Bot.* 68 (2017) 2219–2230, <https://doi.org/10.1093/jxb/erx100>.
- [33] S. De Silva, S.S. Lee, M.B. Dugan, J.L. Anderson, Recent advancements and emerging techniques in nucleic acid isolation, amplification, and detection from diverse complex matrices of human interest, *Trends Anal. Chem.* 185 (2025) 118172, <https://doi.org/10.1016/j.trac.2025.118172>.
- [34] R. Yockteng, A.M.R. Almeida, S. Yee, T. Andre, C. Hill, C.D. Specht, A method for extracting high-quality RNA from diverse plants for next-generation sequencing and gene expression analyses, *Appl. Plant Sci.* 1 (2013), <https://doi.org/10.3732/apps.1300070>.
- [35] A. Becskei, S. Rahaman, The life and death of RNA across temperatures, *Comput. Struct. Biotechnol. J.* 20 (2022) 4325–4336, <https://doi.org/10.1016/J.CSBJ.2022.08.008>.
- [36] B. Ranjbar, P. Gill, Circular dichroism techniques: biomolecular and nanostructural analyses- A review, *Chem. Biol. Drug Des.* 74 (2009) 101–120, <https://doi.org/10.1111/j.1747-0285.2009.00847.x>.
- [37] K.D. Tulsian, S. Jena, M. González-Viegas, R.K. Kar, H.S. Biswal, Structural dynamics of RNA in the presence of choline amino acid based ionic liquid: a spectroscopic and computational outlook, *ACS Cent. Sci.* 7 (2021) 1688–1697, <https://doi.org/10.1021/acscentsci.1c00768>.
- [38] M.T.J. Johnson, E.J. Carpenter, Z. Tian, R. Bruskiewich, J.N. Burris, C.T. Carrigan, M.W. Chase, N.D. Clarke, S. Covshoff, C.W. dePamphilis, P.P. Edger, F. Goh, S. Graham, S. Greiner, J.M. Hibberd, I. Jordon-Thaden, T.M. Kutchan, J. Leebens-Mack, M. Melkonian, N. Miles, H. Myburg, J. Patterson, J.C. Pires, P. Ralph, M. Rolf, R.F. Sage, D. Soltis, P. Soltis, D. Stevenson, C.N. Stewart, B. Surek, C.J. M. Thomsen, J.C. Villarreal, X. Wu, Y. Zhang, M.K. Deyholos, G.K.S. Wong, Evaluating methods for isolating total RNA and predicting the success of sequencing phylogenetically diverse plant transcriptomes, *PLoS One* 7 (2012), <https://doi.org/10.1371/journal.pone.0050226>.